

The Krasnosel'skiĭ-Mann Iterative Method

Chapter 3: The Krasnosel'skiĭ-Mann Iteration

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Notation You'll Need / Recap I

From Chapter 2 (recap). Let $\mathcal{H}$ be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|$. For $T : \mathcal{H} \rightarrow \mathcal{H}$ (or $T : C \rightarrow C$ on a closed convex subset C):

- T is **nonexpansive** if $\|T(x) - T(y)\| \leq \|x - y\|$ for all x, y .
- $\text{Fix}(T) = \{x : T(x) = x\}$ is the set of **fixed points**.
- $\{x_n\}$ **converges weakly** to x^* (written $x_n \rightharpoonup x^*$) if $\langle x_n, z \rangle \rightarrow \langle x^*, z \rangle$ for every $z \in \mathcal{H}$; it converges **strongly** (in norm) if $\|x_n - x^*\| \rightarrow 0$. Strong convergence always implies weak convergence, not conversely (in infinite dimensions).
- The **Picard iteration** is the naive scheme $x_{n+1} = T(x_n)$; it need not converge even when T is nonexpansive and has a fixed point (e.g. a rotation).

Notation You'll Need / Recap II

New in this chapter.

- The **relaxation (or averaging) sequence** $\{\lambda_n\} \subset [0, 1]$ blends the current point x_n with the Picard step $T(x_n)$:

$$x_{n+1} = (1 - \lambda_n)x_n + \lambda_n T(x_n).$$

$\lambda_n = 1$ recovers pure Picard; $\lambda_n = 0$ means “don’t move.”

- The **perturbation term** e_n (or the pair e_n^1, e_n^2) models computational error in evaluating x_n or $T(x_n)$ at each step; it is required to be **summable**, $\sum_n \|e_n\| < +\infty$, so that the accumulated error over infinitely many steps stays finite.
- **Convergence rate** $O(1/\sqrt{n})$ (**plain terms**): the residual that measures how far x_n is from being a fixed point (e.g. $\|x_n - T(x_n)\|$) shrinks no slower than a constant times $1/\sqrt{n}$. Doubling the iteration budget only shrinks the guaranteed error bound by a factor $1/\sqrt{2} \approx 0.71$ — far slower than the geometric (linear) rates you may know from contraction mappings, where doubling the budget can shrink the error by many orders of magnitude. This *sublinear* rate is the price paid for working with mere nonexpansive maps instead of contractions.

Notation You'll Need / Recap III

Running example (used throughout this chapter). Let $\mathcal{H} = \mathbb{R}^2$, $C =$ closed unit disk, and

$$T(x) = \begin{cases} x/\|x\|, & \|x\| > 1, \\ x, & \|x\| \leq 1, \end{cases}$$

the metric projection onto C (firmly nonexpansive, hence nonexpansive, $\text{Fix}(T) = C$). Start at $x_0 = (3, 4)$, $\|x_0\| = 5$. Every iterate stays on the ray through x_0 , so the *unique* point of $\text{Fix}(T)$ that the KM sequence can reach along that ray is

$$x^* = x_0/\|x_0\| = (0.6, 0.8).$$

Writing $x_n = r_n(0.6, 0.8)$, the vector recursion collapses to the scalar recursion (derived in Chapter 1)

$$r_{n+1} = (1 - \lambda_n)r_n + \lambda_n, \quad r_0 = 5.$$

We will plug in constant and decreasing $\{\lambda_n\}$ throughout.

3.1 The KM Iteration: Definition I

Krasnosel'skiĭ-Mann (KM) iteration (3.1)

Let X be a normed space, $C \subset X$ convex, $T : C \rightarrow C$ an operator, and $\{\lambda_n\} \subset [0, 1]$ a real sequence. Starting from any $x_0 \in C$,

$$x_{n+1} = (1 - \lambda_n)x_n + \lambda_n T(x_n) \quad \text{for each } n \geq 0.$$

Symbols: x_n = current iterate; $T(x_n)$ = one Picard step from x_n ; λ_n = how much of the Picard step to take (the *relaxation parameter*).

Intuition. At each step you take a weighted average (a “relaxed” step) between staying put and moving all the way to $T(x_n)$. Mann introduced this in 1953; Krasnosel'skiĭ (1955) studied the special case of *constant* $\lambda_n \equiv \lambda$. Some authors call it the Mann–Toeplitz process, since Toeplitz studied a more general matrix approach of which (3.1) is a special case.

3.1 The KM Iteration: Definition II

Why not just use Picard ($\lambda_n \equiv 1$)? Nonexpansive maps can have rotation-like behaviour with no attraction toward the fixed point, so $x_{n+1} = T(x_n)$ can cycle forever without converging. Averaging with the identity (any $\lambda_n < 1$) damps that oscillation — this is the entire point of the KM scheme.

Running example, constant $\lambda = 0.5$. Using $r_{n+1} = (1 - \lambda)r_n + \lambda$ with $r_0 = 5$:

$$r_0 = 5, r_1 = 3, r_2 = 2, r_3 = 1.5, r_4 = 1.25, r_5 = 1.125, \dots \rightarrow 1.$$

In fact $e_n := r_n - 1$ satisfies $e_{n+1} = (1 - \lambda)e_n$, so $e_n = 4 \cdot 0.5^n$: **geometric** (linear) decay to the fixed point $x^* = (0.6, 0.8)$.

3.1 Mann's Original Matrix Formulation I

Mann's 1953 paper phrased the iteration through an infinite lower-triangular matrix $A = (a_{nk})$ satisfying

- (A1) $a_{nk} \geq 0$ for $n, k \geq 1$ and $a_{nk} = 0$ for $k > n$;
- (A2) $\sum_{k=1}^n a_{nk} = 1$ for each $n \geq 1$ (each row is a convex-combination weight vector).

Generalized Mann process (3.2)

Starting from $x_1 \in C$,

$$v_n = \sum_{k=1}^n a_{nk} x_k, \quad x_{n+1} = T(v_n).$$

Here v_n is a weighted *centroid* of all past iterates $x_1, \dots, x_n$, and the $(n+1)$ -st iterate is the image of that centroid under T . This is denoted briefly (x_1, A, T) ; when $A = I$ (identity) it reduces to plain Picard iteration.

3.1 Mann's Original Matrix Formulation II

Theorem 3.1 (Mann)

Let C be convex compact in a Banach space, $T : C \rightarrow C$ nonexpansive. If either $\{x_n\}$ or $\{v_n\}$ converges, then the other also converges, to the *same* limit, and that limit is a fixed point of T .

3.1 Mann's Original Matrix Formulation III

The Cesàro matrix Λ (a particular A)

$\Lambda_{nk} = \frac{1}{n}$ for $1 \leq k \leq n$. Process (3.3):

$$v_n = \frac{1}{n} \sum_{k=1}^n x_k, \quad x_{n+1} = T(v_n).$$

One checks algebraically that this is equivalent to the *relaxed* recursion (3.4)

$$v_{n+1} = \left(1 - \frac{1}{n+1}\right)v_n + \frac{1}{n+1}T(v_n),$$

i.e. exactly a KM iteration (3.1) with the *decreasing* relaxation sequence $\lambda_n = \frac{1}{n+1}$.

3.1 Mann's Original Matrix Formulation IV

Running example, decreasing $\lambda_n = \frac{1}{n+2}$. (Shifting the index by one to avoid the boundary case $\lambda = 1$ at the very first step.) Using $r_{n+1} = (1 - \lambda_n)r_n + \lambda_n$:

$$r_0 = 5, \quad r_1 = 3, \quad r_2 = 2.333, \quad r_3 = 2, \quad r_4 = 1.8, \quad r_5 = 1.667, \dots$$

One can check directly $e_n = r_n - 1 = \frac{4}{n+1}$ **exactly** — only **sublinear** ($O(1/n)$) decay: much slower than the geometric decay of constant $\lambda = 0.5$, even though both converge.

3.1 Convergence for Continuous Scalar T I

Theorem 3.2 (Mann, [141, Thm. 4])

If $T : [a, b] \rightarrow [a, b]$ is continuous with a *unique* fixed point p , then $\{x_n\}$ from the Cesàro process (3.3) converges to p for every starting point $x_1 \in [a, b]$.

Franks and Marzec later removed the uniqueness assumption. Borwein and Borwein generalized further: let $\{d_n\}$ be nonnegative with $d_1 > 0$, $\sum_{k=1}^{+\infty} d_k = +\infty$, and set $a_{nk} = d_k / \sum_{k=1}^n d_k$.

Theorem 3.3 ([20, Thm. 11])

If $T : [a, b] \rightarrow [a, b]$ has $\text{Fix}(T) \neq \emptyset$, then $\{x_n\}$ from (3.2) converges to a fixed point of T provided **either**

- (a) T is continuous and $\lim_{n \rightarrow \infty} a_{nn} = 0$; **or**
- (b) T is L -Lipschitz and $\limsup_{n \rightarrow \infty} a_{nn} \leq \frac{2}{L+1}$.

3.1 Convergence for Continuous Scalar T II

The Cesàro matrix satisfies (a). *Warning:* Theorem 3.3 fails in higher-dimensional spaces (a counterexample exists).

3.1 Convergence for Continuous Scalar T III

Dotson's “normal Mann process.” Add to (A1)–(A2):

- (A3) $\lim_{n \rightarrow \infty} a_{nk} = 0$ for each fixed k ;
- (A4) $a_{n+1,k} = (1 - a_{n+1,n+1})a_{nk}$ for $1 \leq k \leq n$.

Condition (A4) lets you unwind v_{n+1} as a genuine two-term relaxation:

$$v_{n+1} = a_{n+1,n+1}Tv_n + (1 - a_{n+1,n+1})v_n = \lambda_n T(v_n) + (1 - \lambda_n)v_n,$$

where $\lambda_n := a_{n+1,n+1}$ is read off the *main diagonal* of A . So every normal Mann process is secretly a KM iteration (3.1) with this particular diagonal relaxation sequence. This is why it is sometimes called the “Mann–Dotson” iterate, even though it is a special case of (3.2).

3.1 Relaxed Mean Iteration and Concentrating Arrays I

Combettes and Pennanen generalized (3.2) to an *inexact*, relaxed version for a finite family of operators; the single-operator exact case is:

Relaxed mean iteration scheme (3.5)

$$v_n = \sum_{k=0}^n a_{nk} x_k, \quad x_{n+1} = (1 - \lambda_n) v_n + \lambda_n T(v_n).$$

Reduces to the mean iteration (3.2) when $\lambda_n \equiv 1$.

3.1 Relaxed Mean Iteration and Concentrating Arrays II

Definition 3.1 (Concentrating array, condition (A5))

The array $\{a_{nk}\}$ is *concentrating* if every nonnegative sequence $\{\xi_n\}$ satisfying

$$\xi_{n+1} \leq \sum_{k=0}^n a_{nk} \xi_k \quad \text{for each } n \geq 0$$

converges. *Intuition:* if a nonnegative quantity is bounded above by a weighted average of its own past values under this array, that quantity is forced to settle down — exactly the kind of “monotone-ish” control we will exploit for $\|x_n - p\|$.

3.1 Relaxed Mean Iteration and Concentrating Arrays III

Theorem 3.4 ([46, Thm. 3.5])

Let $\{\lambda_n\} \subset [\delta, 2 - \delta]$ for some $\delta \in (0, 2)$. Suppose $\{a_{nk}\}$ satisfies (A1)–(A3) and (A5), and $T : \mathcal{H} \rightarrow \mathcal{H}$ is **firmly nonexpansive** with $\text{Fix}(T) \neq \emptyset$. Then $\{x_n\}$ from (3.5) **weakly converges** to a fixed point of T .

Combettes and Glaudin later dropped condition (A1) (nonnegativity), moving from a *convex*-hull iteration to an *affine*-hull iteration, under assumptions (B1)–(B4) (boundedness of row sums of absolute values, row sums = 1, columns $\rightarrow 0$, and a concentrating-type condition (3.8) with auxiliary weights χ_n). This allows *negative* entries a_{nk} (Example 3.3), which is genuinely new.

3.1 Relaxed Mean Iteration and Concentrating Arrays IV

Theorem 3.5

Under (B1)–(B4) and $\{\lambda_n\} \subset [\delta, 1 - \delta]$, if

$$\sum_{n=0}^{+\infty} \chi_n \sum_{k=0}^n \sum_{l=0}^k [a_{nk} a_{nl}]^- \|x_k - x_l\|^2 < +\infty,$$

then $\{x_n\}$ from (3.5) converges weakly to a point of $\text{Fix}(T)$.

Open Problem (book). Aside from matrices satisfying Theorem 3.3's hypotheses, is there *any* matrix $\{a_{nk}\}$ for which the mean iteration scheme (3.2) converges?

Python: Section 3.1 — Mann's Cesàro Process

```
1 import numpy as np
2
3 def T(x):
4     """Projection onto closed unit disk in  $R^2$ ."""
5     n = np.linalg.norm(x)
6     return x / n if n > 1 else x
7
8 def cesaro_mann(x1, n_steps):
9     """Mann's original Cesaro process (3.3)-(3.4):
10          $v_{n+1} = (1 - 1/(n+1)) v_n + (1/(n+1)) T(v_n)$ ."""
11     v = np.array(x1, dtype=float)
12     history = [v.copy()]
13     for n in range(1, n_steps + 1):
14         lam = 1.0 / (n + 1)
15         v = (1 - lam) * v + lam * T(v)
16         history.append(v.copy())
17     return np.array(history)
18
19 orbit = cesaro_mann([3.0, 4.0], 8)
20 for k, v in enumerate(orbit):
21     print(f"n={k:2d}  v_n={v}  ||v_n - x*|| = "
22           f"{np.linalg.norm(v - [0.6, 0.8]):.4f}")
23 # Output confirms Theorem 3.1: {x_n} and {v_n} converge to the
24 # same fixed point x* = (0.6, 0.8).
```

3.2 Basic Properties of the KM Sequence I

Let X be a Banach space, $C \subset X$ closed convex, $T : C \rightarrow C$ nonexpansive with $\text{Fix}(T) \neq \emptyset$, and $\{x_n\}$ the KM sequence (3.1).

Basic properties (i)–(ii)

- (i) $\|x_{n+1} - p\| \leq \|x_n - p\|$ for every $n \geq 0$ and every $p \in \text{Fix}(T)$ — **Fejér monotonicity**: the distance to *any* fixed point never increases.
- (ii) $\|x_{n+1} - T(x_{n+1})\| \leq \|x_n - T(x_n)\|$ for every $n \geq 0$ — the *fixed-point residual* is also monotone non-increasing.

Why these are easy but not enough. (i)+(ii) are elementary consequences of nonexpansiveness (short convexity computations). From (ii), being monotone and bounded below by 0, $\|x_n - T(x_n)\|$ *converges to some limit* $c \geq 0$ automatically. But $c = \lim \|x_n - T(x_n)\|$ could in principle be a positive number — we still need the extra fact

$$\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0$$

before we can conclude anything about convergence of $\{x_n\}$ itself. This is the real content of the theorems on this section.

3.2 Basic Properties of the KM Sequence II

Further properties (Borwein et al.)

- (iii) $\|x_{n+1} - x_{n+1}^*\| \leq \|x_n - x_n^*\|$, where $\{x_n^*\}$ is *another* KM sequence for the same T and $\{\lambda_n\}$, started at a different point x_0^* : two KM orbits can only get closer together, never farther apart.
- (iv) If $\limsup_n \lambda_n < 1$ and $\sum_{n=0}^{\infty} \lambda_n = \infty$, then

$$\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = r_C(T) := \inf_{x \in C} \|x - T(x)\|.$$

- (v) If $\sum_{n=0}^{\infty} \lambda_n = \infty$, then

$$\lim_{n \rightarrow \infty} \frac{\|x_{n+1} - x_n\|}{\lambda_n} = \lim_{n \rightarrow \infty} \frac{\|x_0 - x_n\|}{\sum_{k=0}^{n-1} \lambda_k} = r_C(T).$$

Reading (iv): once the step sizes don't decay too fast on average, the residual settles at its *best achievable* value $r_C(T)$ over all of C — which is 0 exactly when $\text{Fix}(T) \neq \emptyset$.

3.2 Reich's Weak Convergence Theorem I

In 1975, Reich gave a simple, far-reaching extension of Mann's original result, for genuinely nonexpansive T (not necessarily firmly nonexpansive) on a suitable Banach space.

Theorem 3.6 (Reich)

Let C be nonempty closed convex in a uniformly convex Banach space X that is Fréchet differentiable (equivalently, one may assume X has the *Opial property* instead). Let $T : C \rightarrow C$ be nonexpansive with a fixed point, $\{\lambda_n\} \subset [0, 1]$ with

$$\sum_{n=0}^{+\infty} \lambda_n(1 - \lambda_n) = +\infty.$$

Then $\{x_n\}$ from (3.1) **converges weakly** to a fixed point of T .

Opial's property: $x_n \rightharpoonup x^*$ implies $\limsup_n \|x_n - x^*\| < \limsup_n \|x_n - x\|$ for all $x \neq x^*$ — i.e. the weak limit is strictly the “closest” point in the limsup sense; every Hilbert space has this property, which is why Theorem 3.6 applies directly to $\mathcal{H}$.

3.2 Reich's Weak Convergence Theorem II

Intuition: why $\sum \lambda_n(1 - \lambda_n) = \infty$? Think of $\lambda_n(1 - \lambda_n)$ as a “useful averaging” indicator, maximized at $\lambda_n = \frac{1}{2}$ and vanishing at $\lambda_n \in \{0, 1\}$:

- If $\lambda_n \rightarrow 0$ *too fast* (e.g. summably), the iteration barely moves each step; the accumulated displacement can run out before the residual is forced to 0 — the sequence can get *permanently stuck* short of the fixed point.
- If $\lambda_n \rightarrow 1$ *too fast* that is actually fine here (it only makes the step closer to plain Picard), but the *condition itself* is written symmetrically in λ_n and $1 - \lambda_n$ because the proof needs a uniformly “non-trivial averaging” effect at every scale it can also fail from the $\lambda_n \rightarrow 1$ side in general operator-theoretic arguments (the symmetric quantity $\lambda_n(1 - \lambda_n)$ is exactly what shows up in the key Fejér-type inequality below).
- In short: *the steps can't be too small AND can't be pure, unrelaxed Picard steps* sustained forever — you need a persistent, non-vanishing amount of averaging.

3.2 Reich's Weak Convergence Theorem III

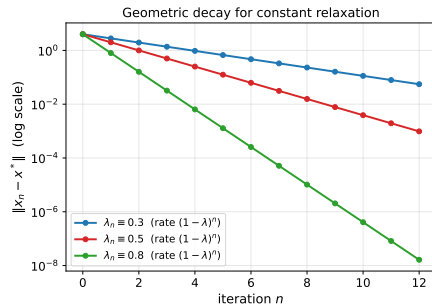
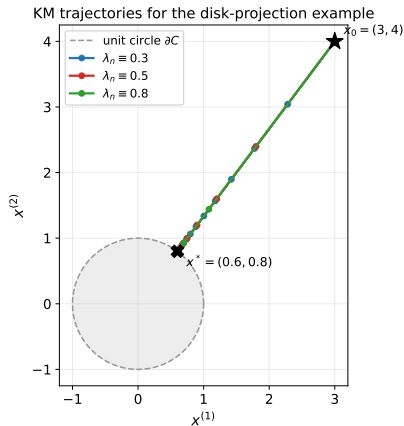
Running example: the condition in action.

- **Good:** $\lambda_n = \frac{1}{n+2}$. Since $\lambda_n(1 - \lambda_n) \sim \frac{1}{n+2}$ for large n , the series $\sum \lambda_n(1 - \lambda_n)$ behaves like the *harmonic series* and **diverges**. Condition holds $\Rightarrow r_n \rightarrow 1$ (converges to x^*), confirmed above: $e_n = 4/(n+1) \rightarrow 0$.
- **Bad:** $\lambda_n = \frac{1}{(n+2)^2}$. Now $\sum \lambda_n < \infty$ (it is a convergent p -series, $p = 2$), so $\lambda_n(1 - \lambda_n) \sim \lambda_n$ and $\sum \lambda_n(1 - \lambda_n) < \infty$ too: condition **fails**. Numerically (running the recursion $r_{n+1} = (1 - \lambda_n)r_n + \lambda_n$):

$$r_0 = 5, \quad r_1 = 3, \quad r_2 = 2.667, \quad r_3 = 2.5, \dots, r_{24} \approx 2.077, \quad \dots, \quad r_{2000} \approx 2.001.$$

The sequence **stalls** near $\|x_n - x^*\| \approx 2$, never reaching $x^* = (0.6, 0.8)$ — exactly as the theory predicts (see Figure on next Python slide).

Figure: Trajectories and Geometric Rate

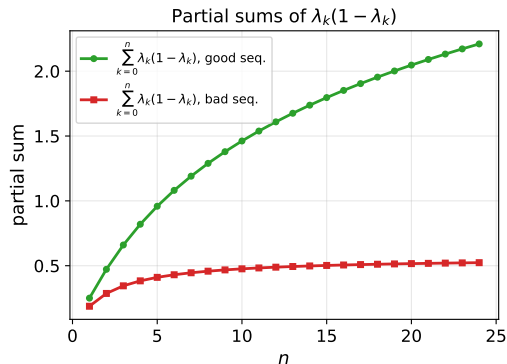
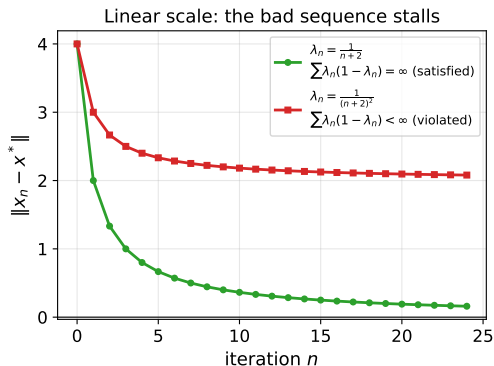


$\|x_n - x^*\|$ vs. n , log scale: straight lines = geometric decay.

Constant $\lambda = 0.3, 0.5, 0.8$: converge to x^* at different speeds.

Figure: Reich's Condition — Satisfied vs. Violated

Reich's condition $\sum \lambda_n(1 - \lambda_n) = \infty$: satisfied vs. violated



Left: with $\lambda_n = \frac{1}{n+2}$ (green) the error keeps shrinking toward 0; with $\lambda_n = \frac{1}{(n+2)^2}$ (red) it stalls near 2. Right: partial sums of $\lambda_k(1 - \lambda_k)$ — one keeps growing (diverges), the other flattens out (converges).

3.2 Step-by-Step: Proof Sketch of the Fejér-Type Estimate I

We derive the key inequality underlying Theorem 3.6 (and reused for Theorem 3.9 later) in a Hilbert space, where the computation is cleanest. Fix $p \in \text{Fix}(T)$.

Step 1 (expand the square). Using $\|(1 - \lambda)a + \lambda b\|^2 = (1 - \lambda)\|a\|^2 + \lambda\|b\|^2 - \lambda(1 - \lambda)\|a - b\|^2$ (the parallelogram-type identity, valid in Hilbert spaces for any $a, b \in \mathcal{H}$, $\lambda \in \mathbb{R}$) with $a = x_n - p$, $b = T(x_n) - p$, $\lambda = \lambda_n$:

$$\|x_{n+1} - p\|^2 = (1 - \lambda_n)\|x_n - p\|^2 + \lambda_n\|T(x_n) - p\|^2 - \lambda_n(1 - \lambda_n)\|x_n - T(x_n)\|^2.$$

Step 2 (use nonexpansiveness). Since $T(p) = p$, $\|T(x_n) - p\| = \|T(x_n) - T(p)\| \leq \|x_n - p\|$, so

$$\|x_{n+1} - p\|^2 \leq \|x_n - p\|^2 - \lambda_n(1 - \lambda_n)\|x_n - T(x_n)\|^2.$$

This is the crucial **quasi-Fejér / Fejér-monotone** inequality: it says $\|x_n - p\|^2$ decreases by *at least* $\lambda_n(1 - \lambda_n)\|x_n - T(x_n)\|^2$ at every step.

3.2 Step-by-Step: Proof Sketch of the Fejér-Type Estimate II

Step 3 (telescope and sum). Summing Step 2's inequality for $n = 0, \dots, N - 1$ telescopes the left side:

$$\sum_{n=0}^{N-1} \lambda_n(1 - \lambda_n) \|x_n - T(x_n)\|^2 \leq \|x_0 - p\|^2 - \|x_N - p\|^2 \leq \|x_0 - p\|^2.$$

Since the right-hand side is bounded uniformly in N , letting $N \rightarrow \infty$ gives

$$\sum_{n=0}^{\infty} \lambda_n(1 - \lambda_n) \|x_n - T(x_n)\|^2 < +\infty.$$

Step 4 (extract a vanishing subsequence). Because $\sum_n \lambda_n(1 - \lambda_n) = +\infty$ by hypothesis, but the weighted sum above with weights $\lambda_n(1 - \lambda_n)$ and terms $\|x_n - T(x_n)\|^2$ is finite, the terms $\|x_n - T(x_n)\|^2$ cannot stay bounded away from 0: we get $\liminf_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0$.

3.2 Step-by-Step: Proof Sketch of the Fejér-Type Estimate III

Step 5 (upgrade liminf to full limit). Combine with property (ii) from before: $\|x_n - T(x_n)\|$ is a *monotone non-increasing* sequence. A monotone sequence with a subsequence going to 0 must itself converge to 0:

$$\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0.$$

Step 6 (weak cluster points are fixed points). $\{x_n\}$ is bounded (by Step 1's monotonicity of $\|x_n - p\|$), so it has weak cluster points; using demiclosedness of $\text{Id} - T$ at 0 (a standard fact for nonexpansive T) and Step 5, every weak cluster point lies in $\text{Fix}(T)$.

Step 7 (Opial argument closes the loop). Fejér monotonicity (Step 1's consequence, property (i)) plus Opial's property forces *all* weak cluster points to coincide $\Rightarrow \{x_n\}$ itself converges weakly to a single point of $\text{Fix}(T)$. ■

3.2 Strong Convergence and Related Results I

Weak convergence is often all one can hope for in infinite dimensions with nonexpansive T ; strong (norm) convergence needs an extra structural assumption on T .

Theorem 3.7 ([195, Thm. 1])

Let X be a real uniformly convex Banach space, $C \subset X$ closed convex, $T : C \rightarrow C$ nonexpansive with $\text{Fix}(T) \neq \emptyset$. Suppose $S := \text{Id} - T$ is φ -**accretive** and $\{\lambda_n\}$ satisfies $\sum_{n=0}^{+\infty} \lambda_n(1 - \lambda_n) = +\infty$. Then $\{x_n\}$ from (3.1) **converges in norm** to a fixed point of T .

(φ -accretivity of $\text{Id} - T$ is a quantitative strengthening of monotonicity that rules out the “flat” directions responsible for merely-weak convergence.) The uniform convexity hypothesis can be relaxed to strict convexity + the Kadec–Klee property, together with demiclosedness of S and $\lambda_n \leq c < 1$, $\sum \lambda_n = +\infty$.

Open Problem (Xu)

For a uniformly convex Banach space X , closed bounded convex C , nonexpansive $T : C \rightarrow C$ with $\text{Fix}(T) \neq \emptyset$, and $\{\lambda_n\} \subset [0, 1]$ with $\sum \lambda_n(1 - \lambda_n) = +\infty$: does the KM sequence $\{x_n\}$ from (3.1) always converge to a fixed point of T (in the sense of Theorem 3.6, but *without* Fréchet differentiability / Opial)?

3.2 Strong Convergence and Related Results II

A continuous-time viewpoint (Boţ & Csetnek). Let $\lambda : [0, +\infty) \rightarrow [0, 1]$ be Lebesgue measurable, $x_0 \in \mathcal{H}$.

Dynamical system (3.10)

$$\dot{x}(t) = \lambda(t)(T(x(t)) - x(t)), \quad x(0) = x_0.$$

Explicit Euler discretization with step size h_n : $x_{n+1} = x_n + h_n \lambda_n (T(x_n) - x_n)$; taking $h_n \equiv 1$ recovers the KM iteration (3.1) exactly. So (3.1) is literally a *time-1 Euler step* of this ODE.

3.2 Strong Convergence and Related Results III

Theorem 3.8

If $\int_0^{+\infty} \lambda(t)(1 - \lambda(t)) dt = +\infty$ **or** $\inf_{t \geq 0} \lambda(t) > 0$, then the unique strong global solution $x(t)$ satisfies $\lim_{t \rightarrow +\infty} (T(x(t)) - x(t)) = 0$ and $x(t)$ converges weakly to a point of $\text{Fix}(T)$, with rate

$$\|T(x(t)) - x(t)\| \leq \frac{d(x_0, \text{Fix } T)}{\sqrt{\underline{\tau} t}}, \quad \underline{\tau} := \inf_{t \geq 0} \lambda(t)(1 - \lambda(t)) > 0,$$

i.e. an $o(1/\sqrt{t})$ (in fact quantitative $O(1/\sqrt{t})$) rate — the continuous-time ancestor of the discrete rates in Section 3.4.

Python: Section 3.2 — Verifying Reich's Condition Numerically

```
1 import numpy as np
2
3 def T(x):
4     n = np.linalg.norm(x)
5     return x / n if n > 1 else x
6
7 def km_orbit(lam_fn, x0, n_steps):
8     x = np.array(x0, dtype=float)
9     hist = [x.copy()]
10    for n in range(n_steps):
11        lam = lam_fn(n)
12        x = (1 - lam) * x + lam * T(x)
13        hist.append(x.copy())
14    return np.array(hist)
15
16 xstar = np.array([0.6, 0.8])
17 good = lambda n: 1.0 / (n + 2)          # sum lam(1-lam) = +infinity
18 bad  = lambda n: 1.0 / (n + 2) ** 2     # sum lam(1-lam) < +infinity
19 for name, lam_fn in [("good", good), ("bad", bad)]:
20     orbit = km_orbit(lam_fn, [3.0, 4.0], 24)
21     err = np.linalg.norm(orbit - xstar, axis=1)
22     psum = sum(lam_fn(n) * (1 - lam_fn(n)) for n in range(24))
23     print(f"{name}: final err={err[-1]:.4f}, sum lam(1-lam)={psum:.4f}")
24 # good: final err -> 0 (Reich's condition holds)
25 # bad : final err -> ~2.08 (condition fails, iteration stalls)
```

3.3 Why Perturbations? I

Motivation. In practice, $T(x_n)$ is rarely evaluated exactly: floating-point error, inexact subroutines, or deliberately injected corrections (e.g. for *superiorization*, or building *inertial* accelerations on top of KM) all add a small error term at each step. We want conditions under which convergence *survives* such perturbations.

Perturbed KM iteration (3.11), Dong et al.

$$x_{n+1} = (1 - \lambda_n)(x_n + e_n^1) + \lambda_n T(x_n + e_n^2) \quad \text{for each } n \geq 0,$$

where $e_n^1, e_n^2 \in \mathcal{H}$ are the perturbations injected into the “keep” branch and the “move” branch respectively.

Summability condition (3.12)

$$\sum_{n=0}^{\infty} \|e_n^i\| < +\infty \quad \text{for each } i = 1, 2.$$

3.3 Why Perturbations? II

Plain terms: the *total* amount of error injected over all (infinitely many) iterations must be finite — individual errors can be nonzero forever, but they must shrink fast enough that their sum converges (e.g. $\|e_n^i\| = O(1/n^{1+\varepsilon})$ works; $\|e_n^i\| = O(1/n)$ does *not*, since $\sum 1/n$ diverges).

3.3 Why Perturbations? III

Intuition: why summable, not just vanishing? If merely $e_n^i \rightarrow 0$ (but not summable, e.g. $\Theta(1/n)$), the errors can still accumulate an *unbounded* total drift, exactly as a random walk with shrinking-but-not-summable steps can still wander off to infinity. Summability guarantees the cumulative nuisance is bounded, so the “clean” Fejér-monotonicity argument of Section 3.2 only picks up a bounded correction term, which does not destroy weak convergence.

3.3 Lemma 3.1: Residual Bound Under Perturbation I

Lemma 3.1

Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be nonexpansive and $\{x_n\}$ generated by (3.11). Then

$$\|x_{n+1} - T(x_{n+1})\| \leq \|x_n - T(x_n)\| + 2(\|e_n^1\| + \|e_n^2\|) \quad \text{for each } n \geq 0.$$

Step-by-step proof.

Step 1. Write out $x_{n+1} - T(x_{n+1})$ using (3.11) and regroup:

$$\begin{aligned} x_{n+1} - T(x_{n+1}) &= (1 - \lambda_n)[(x_n + e_n^1) - T(x_n + e_n^2)] \\ &\quad + [T(x_n + e_n^2) - T((1 - \lambda_n)(x_n + e_n^1) + \lambda_n T(x_n + e_n^2))]. \end{aligned}$$

3.3 Lemma 3.1: Residual Bound Under Perturbation II

Step 2. Apply the triangle inequality, then nonexpansiveness of T to the second bracket (note the argument of the outer T is exactly x_{n+1}):

$$\begin{aligned}\|x_{n+1} - T(x_{n+1})\| &\leq (1 - \lambda_n) \|(x_n + e_n^1) - T(x_n + e_n^2)\| \\ &\quad + \|T(x_n + e_n^2) - T(x_{n+1})\| \\ &\leq (1 - \lambda_n) \|(x_n + e_n^1) - T(x_n + e_n^2)\| \\ &\quad + \lambda_n \|(x_n + e_n^1) - T(x_n + e_n^2)\| + \|e_n^1\| + \|e_n^2\| \\ &= \|(x_n + e_n^1) - T(x_n + e_n^2)\| + \|e_n^1\| + \|e_n^2\|.\end{aligned}$$

3.3 Lemma 3.1: Residual Bound Under Perturbation III

Step 3. Bound the perturbed residual by the clean one, again via nonexpansiveness applied twice:

$$\|(x_n + e_n^1) - T(x_n + e_n^2)\| \leq \|x_n - T(x_n)\| + \|e_n^1\| + \|e_n^2\|.$$

Step 4. Combine Steps 2–3:

$$\|x_{n+1} - T(x_{n+1})\| \leq \|x_n - T(x_n)\| + 2(\|e_n^1\| + \|e_n^2\|). \quad \blacksquare$$

Reading it: the fixed-point residual can only grow by (twice) the size of the perturbations injected at that step — so under the summability condition (3.12), the residual's growth over all time is bounded, which is exactly the leverage used in Theorem 3.9's proof.

3.3 Theorem 3.9: Weak Convergence with Perturbations I

Theorem 3.9

Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be nonexpansive with $\text{Fix}(T) \neq \emptyset$. Assume $\sum_{n=0}^{\infty} \lambda_n(1 - \lambda_n) = \infty$. Then $\{x_n\}$ generated by the perturbed KM iteration (3.11) **weakly converges** to a point in $\text{Fix}(T)$.

Proof sketch (mirrors Section 3.2, absorbing the perturbation).

Step 1. Expand $\|x_{n+1} - p\|^2$ using the same Hilbert identity as before, then bound each nonexpansive image using the elementary inequality $\|y + e\| \leq (1 + \|e\|) \|y\| + \|e\|$ (*Lemma 2.9* in the book), to control the effect of e_n^1, e_n^2 :

$$\begin{aligned} \|x_{n+1} - p\|^2 &\leq (1 + \|e_n^1\| + \|e_n^2\|) \|x_n - p\|^2 + (\|e_n^1\| + \|e_n^2\| + \|e_n^1\|^2 + \|e_n^2\|^2) \\ &\quad - \lambda_n(1 - \lambda_n) \|(x_n + e_n^1) - T(x_n + e_n^2)\|^2. \end{aligned}$$

3.3 Theorem 3.9: Weak Convergence with Perturbations II

Step 2. Because $\sum_n (\|e_n^1\| + \|e_n^2\|) < \infty$ by (3.12), a discrete Gronwall-type lemma (*Lemma 2.8*) applied to the recursion of Step 1 shows $\lim_n \|x_n - p\|$ **exists** (hence $\{x_n\}$ is bounded), *despite* the multiplicative $(1 + \|e_n^1\| + \|e_n^2\|)$ factor — summability keeps the cumulative inflation finite.

Step 3. Rearranging Step 1 and summing over n (as in Section 3.2's Step 3) gives

$$\sum_{n=0}^{\infty} \lambda_n (1 - \lambda_n) \|(x_n + e_n^1) - T(x_n + e_n^2)\|^2 < +\infty,$$

so, since $\sum_n \lambda_n (1 - \lambda_n) = \infty$, $\liminf_n \|(x_n + e_n^1) - T(x_n + e_n^2)\| = 0$.

Step 4. Using nonexpansiveness once more, $\|x_n - T(x_n)\| \leq \|(x_n + e_n^1) - T(x_n + e_n^2)\| + \|e_n^1\| + \|e_n^2\|$, so $\liminf_n \|x_n - T(x_n)\| = 0$ too (the $e_n^i \rightarrow 0$ since summable).

3.3 Theorem 3.9: Weak Convergence with Perturbations III

Step 5. Combine Step 4 with Lemma 3.1 (the residual can only grow by a *summable* amount each step) to upgrade the liminf to a genuine limit:

$$\lim_{n \rightarrow \infty} \|x_n - T(x_n)\| = 0.$$

Step 6. $\{x_n\}$ bounded (Step 2) $\Rightarrow$ nonempty set $\omega_w(x_n)$ of weak cluster points; demiclosedness of $\text{Id} - T$ plus Step 5 give $\omega_w(x_n) \subseteq \text{Fix}(T)$.

Step 7. A (perturbed) Opial-type uniqueness-of-weak-limit argument forces $\omega_w(x_n)$ to be a single point $\Rightarrow \{x_n\}$ converges weakly to a point of $\text{Fix}(T)$. ■

Takeaway: the same relaxation condition $\sum \lambda_n(1 - \lambda_n) = \infty$ that drives clean convergence *also* suffices with summable perturbations — the KM scheme is robust to small, well-controlled errors.

3.3 Running Example: A Concrete Perturbed Iteration I

Take $\lambda_n \equiv 0.5$ and the perturbation $e_n = (\frac{0.01}{n+1}, 0)$ (added once per step, i.e. the simpler inexact form $x_{n+1} = (1 - \lambda_n)x_n + \lambda_n(T(x_n) + e_n)$, cf. Liang et al.'s scheme (3.25) discussed in Section 3.4).

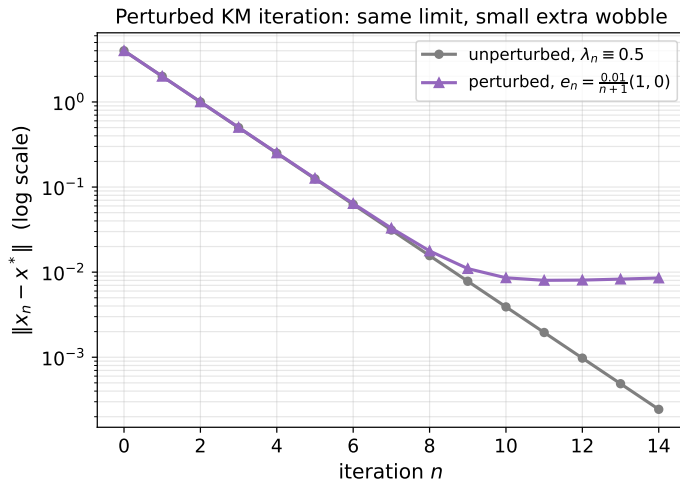
n	x_n (perturbed)	$\ x_n - x^*\ $ (perturbed)	$\ x_n - x^*\ $ (clean)
0	(3.000, 4.000)	4.000	4.000
1	(1.805, 2.400)	2.003	2.000
2	(1.206, 1.600)	1.003	1.000
3	(0.905, 1.199)	0.503	0.500
4	(0.755, 0.999)	0.252	0.250
5	(0.680, 0.898)	0.127	0.125
6	(0.643, 0.848)	0.064	0.0625

The perturbed iterates track the clean geometric decay $4 \cdot 0.5^n$ very closely, with a small persistent offset from the $(0.01/(n+1), 0)$ nudges in the first coordinate — and still converge to $x^* = (0.6, 0.8)$.

3.3 Running Example: A Concrete Perturbed Iteration II

A caveat worth stating precisely. The sequence $0.01/(n+1)$ used above is illustrative over a short horizon, but $\sum_n 0.01/(n+1)$ actually **diverges** (harmonic series) — it is *not* summable, so it does not strictly satisfy hypothesis (3.12) of Theorem 3.9! Over the 6 steps shown, the geometric contraction from $\lambda = 0.5$ dominates the slowly-growing error and convergence looks fine numerically. To have a perturbation that *rigorously* satisfies (3.12) (guaranteeing convergence for *all* n , not just the first few), replace it with, e.g., a genuinely summable choice such as $e_n = (\frac{0.01}{(n+1)^2}, 0)$, since $\sum_n 1/(n+1)^2 = \pi^2/6 < \infty$. This distinction — summable versus merely vanishing — is exactly the content of condition (3.12) and is easy to overlook in practice.

Figure: Perturbed vs. Clean Iteration



Perturbed KM iteration ($\lambda \equiv 0.5$, $e_n = \frac{0.01}{n+1}(1, 0)$) tracks the clean geometric decay closely and reaches the same fixed point $x^* = (0.6, 0.8)$.

3.3 Bounded Perturbation Resilience (BPR) and Superiorization I

Motivation. If an algorithm is known to be robust to *any* sufficiently small, summable perturbation, one can deliberately inject perturbations that steer the iterates toward a *secondary* objective (e.g. regularization) without destroying convergence to the original solution set — this is the *superiorization* methodology.

3.3 Bounded Perturbation Resilience (BPR) and Superiorization II

Definition 3.2 (Bounded Perturbation Resilience, BPR)

Let $\Theta \subseteq \mathcal{H}$ and let $\mathbf{A}_\Psi : \Theta \rightarrow \mathcal{H}$ be an algorithmic operator such that the **basic algorithm** (3.20)

$$x_{n+1} = \mathbf{A}_\Psi(x_n), \quad x_0 \in \Theta,$$

converges to a point in a solution set Ψ . $\mathbf{A}_\Psi$ is **bounded perturbations resilient** if, for *any* $y_0 \in \Theta$, the perturbed sequence

$$y_{n+1} = \mathbf{A}_\Psi(y_n + \rho_n v_n), \quad n \geq 0,$$

also converges to a point of Ψ , provided:

- (a) $\{v_n\}$ is bounded;
- (b) $\{\rho_n\} \subset \mathbb{R}$ is positive with $\sum_n \rho_n < \infty$;
- (c) $y_n + \rho_n v_n \in \Theta$ for each $n \geq 0$ (only needed when $\Theta \neq \mathcal{H}$).

3.3 Bounded Perturbation Resilience (BPR) and Superiorization III

Applying this to KM. Take $\mathbf{A}_\Psi^n = (1 - \lambda_n) \text{Id} + \lambda_n T$ (the right-hand side of (3.1) itself). Its bounded perturbation (3.22) is

$$x_{n+1} = \mathbf{A}_\Psi^n(x_n + \rho_n v_n) = (1 - \lambda_n)(x_n + \rho_n v_n) + \lambda_n T(x_n + \rho_n v_n).$$

Setting $e_n^1 = e_n^2 = \rho_n v_n$ turns this into exactly the perturbed scheme (3.11), and $\sum_n \rho_n < \infty$ plus boundedness of $\{v_n\}$ gives $\sum_n \|e_n^i\| < \infty$: condition (3.12) holds automatically!

Theorem 3.10

Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be nonexpansive with $\text{Fix}(T) \neq \emptyset$ and $\sum_{n=0}^{\infty} \lambda_n(1 - \lambda_n) = \infty$. Then $\{x_n\}$ from the bounded-perturbation scheme (3.22) weakly converges to a point of $\text{Fix}(T)$.

Remark 3.1. (1) The KM iteration for nonexpansive operators is therefore bounded perturbations resilient — a direct corollary of Theorem 3.9. (2) This BPR property is precisely the “certificate” needed to justify superiorized/inertial variants of KM built on top of the basic scheme.

Python: Section 3.3 — Perturbed KM and Summability

```
1 import numpy as np
2
3 def T(x):
4     n = np.linalg.norm(x)
5     return x / n if n > 1 else x
6
7 def perturbed_km(x0, n_steps, lam=0.5, err_power=1.0, err_scale=0.01):
8     """ $x_{n+1} = (1-lam) x_n + lam (T(x_n) + e_n)$ ,
9          $e_n = err\_scale / (n+1)**err\_power * (1, 0)$ ."""
10    x = np.array(x0, dtype=float)
11    hist = [x.copy()]
12    for n in range(n_steps):
13        e_n = np.array([err_scale / (n + 1) ** err_power, 0.0])
14        x = (1 - lam) * x + lam * (T(x) + e_n)
15        hist.append(x.copy())
16    return np.array(hist)
17
18 xstar = np.array([0.6, 0.8])
19 # NOT summable: err_power=1.0 (sum 1/(n+1) diverges) -- (3.12) violated
20 orbit_bad = perturbed_km([3.0, 4.0], 6, err_power=1.0)
21 # summable: err_power=2.0 (sum 1/(n+1)^2 = pi^2/6 < infinity) -- (3.12) holds
22 orbit_good = perturbed_km([3.0, 4.0], 6, err_power=2.0)
23 for n in range(7):
24     print(n, np.linalg.norm(orbit_bad[n]-xstar),
25           np.linalg.norm(orbit_good[n]-xstar))
```

3.4 Why Rates Matter, and the Lipschitz Case I

Motivation. Weak/strong convergence tells us $\{x_n\}$ *eventually* reaches $\text{Fix}(T)$, but says nothing about *how many* iterations are needed for a target accuracy. Convergence rate results quantify this.

Theorem 3.11 (He et al.)

Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be L -Lipschitz with $-T$ monotone, and $\lambda_n \equiv \lambda$ with $\max\left\{\frac{L^2 - 1}{L^2 + 1}, 0\right\} < \lambda < 1$. Then, with x^* the unique fixed point,

$$\|x_n - x^*\| \leq \frac{[\lambda^2 + (1 - \lambda)^2 L^2]^{n/2}}{1 - \sqrt{\lambda^2 + (1 - \lambda)^2 L^2}} \|x_1 - x_0\|.$$

Choosing the *optimal* $\lambda = \frac{L^2}{L^2 + 1}$ gives the (essentially) optimal geometric rate

$$\|x_n - x^*\| \leq \sqrt{L^2 + 1} (1 + \sqrt{L^2 + 1}) \left(\frac{L}{\sqrt{L^2 + 1}} \right)^n \|x_1 - x_0\|.$$

3.4 Why Rates Matter, and the Lipschitz Case II

Reading it: under this extra Lipschitz/monotonicity structure, KM converges *geometrically* (linearly) — our constant- λ running example already showed this behaviour, but Theorem 3.11 makes the base of the geometric rate explicit in terms of L and the optimal choice of λ .

3.4 Why Rates Matter, and the Lipschitz Case III

The harder, general case. For a merely nonexpansive T (no Lipschitz constant beyond 1, no extra monotonicity), there is in general **no** rate on $\|x_n - x^*\|$ itself. The workhorse quantity instead is the **fixed-point residual** $\|x_n - T(x_n)\|$, whose convergence to 0 is called **asymptotic regularity** (recall this quantity is exactly the one shown monotone non-increasing in property (ii), Section 3.2). Goebel and Kirk showed: for every $\varepsilon > 0$, $\|x_n - T(x_n)\| \leq \varepsilon$ for all $n \geq n_0(\varepsilon, C)$, with n_0 independent of x_0 and T (later refined so n_0 depends on C only through its diameter).

3.4 The Sharp $O(1/\sqrt{n})$ Rate I

Baillon and Bruck first showed, for constant $\lambda_n \equiv \lambda$ on a bounded C , the rate $\|x_n - T(x_n)\| = O(1/\log n)$, later conjecturing a universal constant κ with

$$\|x_n - T(x_n)\| \leq \kappa \frac{\text{diam}(C)}{\sqrt{\sum_{i=0}^n \lambda_i(1 - \lambda_i)}}. \quad (3.23)$$

Reading (3.23) in plain terms: the fixed-point residual is controlled by $1/\sqrt{\cdot}$ of the *same* quantity $\sum \lambda_i(1 - \lambda_i)$ from Reich's condition. This is precisely what " $O(1/\sqrt{n})$ " means here: if $\sum_{i=0}^n \lambda_i(1 - \lambda_i)$ grows proportionally to n (e.g. constant λ , so each term contributes a fixed positive amount), the residual shrinks like a constant divided by $\sqrt{n}$ — doubling n shrinks the guaranteed bound by only $1/\sqrt{2}$. For affine T and $\lambda_n \equiv 1/2$: tight bound with $\kappa = 1/\sqrt{2\pi}$; for general λ_n , optimal constant $\kappa = \max_{x \geq 0} \sqrt{x} e^{-x} I_0(x) \sim 0.4688$ (I_0 : modified Bessel function of the first kind).

3.4 The Sharp $O(1/\sqrt{n})$ Rate II

Theorem 3.12 (Bravo & Cominetti, [26, Thm. 1.1])

The constant $\kappa = 1/\sqrt{\pi}$ in (3.23) is **tight**: for every $\kappa < 1/\sqrt{\pi}$ there exist a nonexpansive T on the unit cube $C = [0, 1]^{\mathbb{N}} \subseteq \ell^\infty(\mathbb{N})$, an initial point $x_0 \in C$, and a constant $\lambda_n \equiv \lambda$, such that for some $n \geq 0$,

$$\|x_n - T(x_n)\| > \kappa \frac{\text{diam}(C)}{\sqrt{\sum_{i=0}^n \lambda_i(1 - \lambda_i)}}.$$

So (3.23) with $\kappa = 1/\sqrt{\pi}$ is the best possible constant that can hold for *all* nonexpansive maps and constant relaxations — no universal improvement is possible.

Theorem 3.13 (Matsushita, [148, Thm. 3.1])

Let $C \subset \mathcal{H}$ be closed, $T : C \rightarrow C$ nonexpansive with $\text{Fix}(T) \neq \emptyset$, $\{x_n\}$ from (3.1) with $\{\lambda_n\} \subset [0, 1]$. If $\sigma_n := \sum_{j=0}^n \lambda_j(1 - \lambda_j) \rightarrow \infty$, then

$$\|x_n - T(x_n)\| = o(1/\sqrt{\sigma_n}), \quad \text{i.e.} \quad \lim_{n \rightarrow \infty} \sqrt{\sigma_n} \|x_n - T(x_n)\| = 0.$$

3.4 The Sharp $O(1/\sqrt{n})$ Rate III

This upgrades the big- O bound (3.23) to **little-o**: not only bounded by a constant times $1/\sqrt{\sigma_n}$, but the product actually vanishes in the limit.

3.4 Complexity Bounds for the Perturbed Iteration I

Liang et al. studied the inexact iteration

$$x_{n+1} = (1 - \lambda_n)x_n + \lambda_n(T(x_n) + e_n), \quad n \geq 1. \quad (3.25)$$

and Dong et al. extended their pointwise/ergodic bounds to the fully perturbed scheme (3.11) (which contains (3.25) as the special case $e_n^1 = 0$, $e_n^2 = e_n$).

Setup. Let $\bar{\tau} = \sup_n \lambda_n(1 - \lambda_n)$, $\underline{\tau} = \inf_n \lambda_n(1 - \lambda_n)$, $d_1 = d(x_1, \text{Fix}(T))$, and define bounded auxiliary constants v_1, v_2, v_3 collecting suprema of residuals and perturbation norms (full definitions in (3.11)-preceding text).

Theorem 3.14 (Pointwise Iteration-Complexity Bound)

If $0 < \inf_n \lambda_n \leq \sup_n \lambda_n < 1$ and $\{(n+1)(\|e_n^1\| + \|e_n^2\|)\} \in \ell_+^1$ (i.e. that weighted series converges), then, with C_1 a finite constant built from v_1, v_2, v_3 and the perturbation series,

$$\|x_n - T(x_n)\| \leq \sqrt{\frac{d_1^2 + C_1}{\underline{\tau}(n+1)}}.$$

3.4 Complexity Bounds for the Perturbed Iteration II

This is precisely an $O(1/\sqrt{n})$ pointwise rate for the residual, *including* the effect of summable perturbations (through C_1).

3.4 Complexity Bounds for the Perturbed Iteration III

Theorem 3.15 (Ergodic Iteration-Complexity Bound)

With $\Lambda_n = \sum_{j=0}^n \lambda_j$ and $C_2 = \sum_{n=0}^{\infty} [(1 - \lambda_n) \|e_n^1\| + \lambda_n \|e_n^2\|] < \infty$,

$$\left\| \frac{1}{\Lambda_n} \sum_{j=0}^n \lambda_j (x_j - T(x_j)) \right\| \leq \frac{2(d_1 + C_2)}{\Lambda_n}.$$

In particular, if $\inf_n \lambda_n > 0$, then $\Lambda_n = \Theta(n)$ and the ergodic (weighted-average) residual is $O(1/n)$ — *faster* than the $O(1/\sqrt{n})$ pointwise rate, since averaging cancels oscillations.

3.4 Complexity Bounds for the Perturbed Iteration IV

Extending to inexact $T(x_n)$ in Banach spaces. Bravo et al. established, for $\tau_n = \sum_{k=0}^n \lambda_k(1 - \lambda_k)$ and $\sigma(y) = \min\{1, 1/\sqrt{\pi y}\}$:

Theorem 3.16 ([27, Thm. 3]) — setup

X Banach, C closed convex, $T : C \rightarrow C$ nonexpansive, $\{x_n\}$ from the inexact scheme (3.25) with the boundedness condition $x_n \in C$, $\|T(x_n) - x_0\| \leq \kappa$ for all n .

If additionally $\sum_n \|e_n\| < \infty$ and λ_n is bounded away from 0 and 1, there is a constant $c \geq 0$ with

$$\|x_n - T(x_n)\| \leq \frac{c}{\sqrt{n}} + \sum_{i \geq \lfloor n/2 \rfloor} 2 \|e_i\|.$$

More generally, for nondecreasing φ with $\mu = \sum_n \varphi(n) \|e_n\| < \infty$:

$$\|x_n - T(x_n)\| \leq \frac{c}{\sqrt{n}} + \frac{2\mu}{\varphi(\lfloor n/2 \rfloor)}.$$

3.4 Complexity Bounds for the Perturbed Iteration V

Rate depends on the error's decay rate. If $\|e_n\| = O(1/n^a)$ for $a \geq \frac{1}{2}$ and λ_n bounded away from 0, 1:

- if $\frac{1}{2} \leq a < 1$: $\|x_n - T(x_n)\| = O(1/n^{a-1/2})$;
- if $a = 1$: $\|x_n - T(x_n)\| = O(\log n / \sqrt{n})$;
- if $a > 1$: $\|x_n - T(x_n)\| = O(1/\sqrt{n})$ (error decay no longer the bottleneck; recovers the unperturbed rate).

A sobering closing remark. All rates obtained so far for nonexpansive T are only **sublinear**: fast early progress, then a slow tail. Oblomskaja constructed a (linear) example where convergence is slower than $n^{-\tau}$ for every $\tau \in (0, 1)$ — KM convergence for nonexpansive maps can be arbitrarily slow in the worst case. Relaxation strategies and inertial acceleration (Chapters 5–7) are the main remedies developed for this.

Python: Section 3.4 — Empirical $O(1/\sqrt{n})$ Rate

```
1 import numpy as np
2
3 def T(x):
4     n = np.linalg.norm(x)
5     return x / n if n > 1 else x
6
7 def residual_orbit(lam_fn, x0, n_steps):
8     """Track the fixed-point residual ||x_n - T(x_n)|| along the KM orbit."""
9     x = np.array(x0, dtype=float)
10    residuals = []
11    sigma = 0.0 # running sum_{k<=n} lambda_k(1-lambda_k)
12    sigmas = []
13    for n in range(n_steps):
14        lam = lam_fn(n)
15        residuals.append(np.linalg.norm(x - T(x)))
16        sigma += lam * (1 - lam)
17        sigmas.append(sigma)
18        x = (1 - lam) * x + lam * T(x)
19    return np.array(residuals), np.array(sigmas)
20
21 res, sig = residual_orbit(lambda n: 0.5, [3.0, 4.0], 40)
22 bound = 1.0 / np.sqrt(np.pi) * 2.0 / np.sqrt(np.maximum(sig, 1e-12)) # (3.23)-style
23 for n in [0, 5, 10, 20, 39]:
24     print(f"n={n:3d} residual={res[n]:.6f} 1/sqrt(pi*sigma_n) bound={bound[n]:.6f}")
25 # Confirms: residual stays comfortably below the O(1/sqrt(sigma_n)) envelope.
```

Chapter 3 Summary I

- **3.1** The KM iteration $x_{n+1} = (1 - \lambda_n)x_n + \lambda_n T(x_n)$ generalizes Picard iteration via relaxation; Mann's matrix formulation (3.2) and its Cesàro special case recover it as a diagonal-relaxation scheme (Dotson's normal Mann processes); Combettes–Pennanen/Glaudin push this to general (even sign-indefinite) concentrating arrays.
- **3.2** The basic Fejér-monotonicity properties (i)-(ii) are easy but insufficient; Reich's Theorem 3.6 ($\sum \lambda_n(1 - \lambda_n) = \infty$) gives weak convergence for nonexpansive T in Hilbert/Opial spaces; strong convergence (Theorem 3.7) needs φ -accretivity of $\text{Id} - T$.
- **3.3** Perturbed KM (3.11) with summable errors ($\sum \|e_n^i\| < \infty$) preserves weak convergence (Theorem 3.9) under the *same* condition on $\{\lambda_n\}$; this underlies bounded perturbation resilience (Definition 3.2) and superiorization.
- **3.4** Rates are sublinear in general: the sharp universal bound $\|x_n - T(x_n)\| \leq \frac{1}{\sqrt{\pi}} \frac{\text{diam}(C)}{\sqrt{\sum_{i \leq n} \lambda_i(1 - \lambda_i)}}$ (Bravo–Cominetti) is tight; perturbed/inexact versions cost an extra term controlled by the decay rate of $\{e_n\}$; worst-case convergence can be slower than any polynomial rate.

Chapter 3 Summary II

Running example, one-line recap. On the unit-disk projection with $x_0 = (3, 4)$: constant λ gives geometric decay $4(1 - \lambda)^n$; $\lambda_n = \frac{1}{n+2}$ gives sublinear $4/(n+1)$ decay (condition satisfied); $\lambda_n = \frac{1}{(n+2)^2}$ *stalls* near $\|x_n - x^*\| \approx 2$ (condition violated); a small summable perturbation barely disturbs convergence to $x^* = (0.6, 0.8)$.